MILITARY SPECIFICATION

PROTECTORS, ORTHOPEDIC CAST

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope This specification covers plastic waterproof protectors for orthopedic casts of the arm and leg (see 6.1)

1.2 Classification Protectors covered by this specification shall be of two types, as follows

Type I - Arm cast protector

Type II - Leg cast protector.

2 APPLICABLE DOCUMENTS

2.1 Issues of documents The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

L-P-375 - Plastic Film, Flexible, Vinyl Chloride.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-406 - Plastic, Methods of Testing
FED-STD-601 - Rubber, Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage.

3. REQUIREMENTS

3.1 Material

3.1.1 Vinyl film The film or sheet shall be clear or translucent polyvinyl chloride conforming to L-P-375, Type I, Class 1.

3.1.1.1 Thickness Thickness of the film used for the arm cast cover shall be 0.0060 ± 0.0006 inch and for the leg cast cover shall be 0.0080 ± 0.0008 inch

3.1.2 Rubber for belts Rubber for belts shall be pure gum rubber with the properties listed in table I when tested as specified in section 4.

TABLE I Rubber properties

Property	Value
Thickness	13-20 mils.
Tensile strength, initial, psi, min	2000
Ultimate elongation, initial (percent)	500-600
Tensile stress at 100 percent elongation psi maximum	125

3.2 Design and construction The arm and leg protectors shall be constructed of vinyl plastic (see 3.1.1) The plastic film shall be fabricated into a bag by the thermo-welding (heat seal) or high frequency (electric) process, with a press-grip type fastener sealed to one side of bag The protector shall be furnished with a rubber belt (3.1.2) which has a matching fastener bonded to one end for attaching, by finger pressure, to the bag fastener. The arm and leg protectors shall be of the design and dimensions shown in figure 1.

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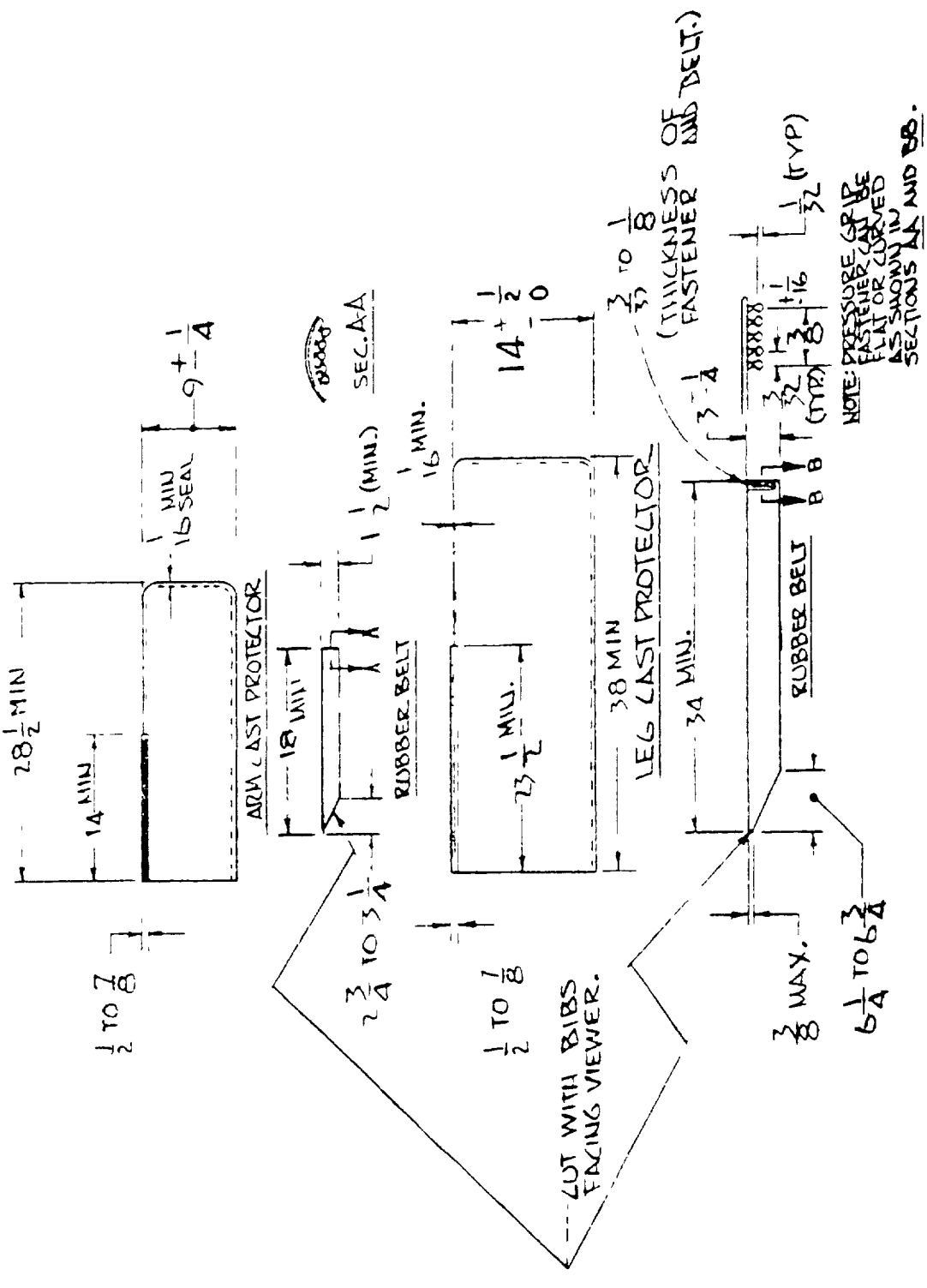


FIGURE 1. Protector, orthopedic cast.

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3.2.1 Sealed joint and adhesive joint tests. The thermal and electronically sealed joints shall not separate when tested as specified in 4.4 3 The adhesive joint shall not separate by more than 1/16 inch when tested as specified in 4.4 3 1

3.2.1.1 Fastener test. The press-grip fastener of the arm or leg protector shall be meshed with the press grip fastener of the corresponding belt and tested as specified in 4.4 4 The meshed joint shall not separate.

3.3 Instructions for use Each orthopedic cast protector shall be supplied with an instructions for use sheet outlining the use of the item. The following or similar statements must be included in the instructions for use "Patients with varicose veins must obtain a doctor's advice before using this product
Not recommended for use in tub after using in shower
Not recommended for use with open wounds or injuries subject to infection, although additional safety may be obtained by wrapping a towel around leg or arm at top of cast or bandage before placing protector."

3.4 Workmanship Workmanship shall be first class throughout. The finished protectors and belts shall be free from all defects which will detract from their appearance or may impair their serviceability

4. QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1 1 Records Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate

4.1.2 Inspection Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the vinyl film and rubber belting may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be domonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements

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4.2 Sampling

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table II The unit of product for examination purposes shall be one (1) protector.

TABLE II Sampling criteria for examination.

Type of defect	Inspection level	AQL (Percent defective)
Major	II	1.5
Minor	II	4.0
Dimensional	S-3	4.0

4 2 2 For test of materials Sampling for tests of materials shall be in sufficient quantity from each lot to perform all tests specified herein.

4 2 2 1 For tests of end item Sampling for tests of the end item shall be conducted in accordance with MIL-STD-105, and as indicated in table III. Unit of product for test purposes shall be one (1) protector

TABLE III Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection Level	AQL
Vinyl film properties	3.1.1	4 4	S-3	0.65
Rubber properties	3 1.2	4.4	S-3	0.65
Heat seal and adhesive joint test	3 2.1	4 4	S-3	0.65
Press-grip fastener test	3 2.1.1	4 4	S-3	0.65

4 3 Examination of the end item The protector shall be examined for, but not restricted to, the defects listed in table IV. Additional defects shall be classified in accordance with MIL-STD-105.

TABLE IV Classification of defects

Category	Defect
Major	
101	Fastener not completely attached to protector
102	Press-grip fastener does not work properly.
103	Heat seal uneven with visible skips.
104	Tears or holes.

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TABLE IV. Classification of defects. - Continued

Category	Defect
Minor	
201	Cover or belt not clean (not free of oil, dirt or extraneous matter)
202	Uneven and jaggec edges of cover or belt.

4.3 1 Dimensional examination Any dimension not within the specified tolerance shall be classified as a defect.

4 3.2 Packaging inspection The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Vinyl film. The vinyl film shall be tested as described in the referenced subsidiary specification.

4.4 2 Rubber belting The rubber belting shall be tested by the method shown in table V.

TABLE V Test methods (rubber belting).

Characteristic	Requirement paragraph	Test Method FED-STD-601
Thickness	3 1 2	2011
Tensile strength	3 1 2	4111
Ultimate elongation	3 1 2	4121
Tensile strength at 100 percent elongation	3 1 2	4131

4 4.3 Parting test, seals The seals shall be tested manually for parting

4 4 3 1 Parting test, adhesive joints, rubber belts Samples shall be kept in water at 120° F minimum for 1/2 hour Samples shall be cooled to room temperature before testing Mesh the fastener on the bag with the fastener on the belt, with the belt pointing away from the bag Hold the fastener on the belt firmly to the fastener on the bag Stretch the 1 inch of rubber next to the fastener to 4 inches for a minimum of 5 seconds. Failure of joint shall not be greater than 1/16 inch.

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4.4.4 Testing of "press-grip" fastener. The "press-grip" fastener of the belt shall be firmly meshed with the "press-grip" fastener of the arm or leg protector. The "press-grip" fastener of the protector shall be held firmly or fastened down. The belt shall be pulled in a direction parallel to the width of the "press-grip" fasteners, with the belt pointing away from the protector.

5 PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue One package containing twelve protectors of the type specified, constitutes one unit of issue

5 1.2 Level A

5 1.2.1 Individual package. One protector along with sealing and retaining belts shall be neatly rolled or folded and packaged along with instructions for use in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5 1 2.2 Unit package Twelve individual packages containing protectors of one type only shall be packaged in a box conforming to PPP-B-636, class domestic Closure shall be as specified in the appendix of the box specification

5 1 3 Level C Protectors shall be clean, dry and individually packaged. Twelve individually packaged protectors of one type only shall be packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing Packing shall be level A, B or C, as specified (see 6.2)

5 2 1 Level A Twelve unit packages containing type I protectors or six unit packages containing type II protectors shall be packaged in a box designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification

5 2 2 Level B Twelve unit packages containing type I protectors or six unit packages containing type II protectors shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic Closure shall be as specified in the appendix of the box specification.

5 2 3 Level C The packaged protectors shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity These packs shall conform to the applicable carrier's rules and regulations

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5.3 Marking.

5.3.1 Individual package. In addition to any commercial markings, each package shall be marked to identify contents and quantity contained

5.3.2 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. Contract or Purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cast protectors are plastic waterproof covers suitable for use during tub or shower baths to protect arm or leg casts, bandages, surgical wounds or skin problems.

6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.
b. Type (see 1.2)
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. National Stock Number.

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog

National Stock Number	Item Identification
6515-00-142-8991	PROTECTOR, ORTHOPEDIC CAST, Arm, 12s.
6515-00-142-8992	PROTECTOR, ORTHOPEDIC CAST, Leg, 12s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6515-2056

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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